Biochemistry Cume – January 27, 2012

1. (24 points) Draw the chemical structure of any tripeptide that contains one methionine residue and has a net charge of –1 at neutral pH.

2. (6 points) Name the peptide that you have drawn in problem 1.

3. (24 points) Draw a reasonably accurate titration curve for your peptide from problem one. Be sure to label the axes appropriately and accurately.

4. (6 points) The peptide from problem one is broken into its constituent amino acids. Suggest a method for doing this.

5. (3 points) Consider solutions that contain (a) 1 gram per liter of the peptide from problem one, or (b) 1 gram per liter of the mixture of amino acids from problem four. Which solution is more strongly buffered against changes in pH?

6. (17 points) Justify your answer to question 5.

7. (10 points) What products would result from the reaction of the peptide from problem one with phenylisothiocyanate?

8. (10 points) Draw the predominant chemical structure at pH 2 for one of the products from problem 7.